Setting up Apache Hive

Learning objectives

By the end of this lab session, you will be able to:

- Install and configure Hive on a Hadoop cluster.
- Start and stop Hive services and ensure they are running.
- Launch Hive CLI or Beeline.

Prerequisites

Ensure you have:

- Apache Hadoop cluster with one master and two workers.
- MySQL server with a user `hive` and password `hive`.

Solution – step-by-step

1. Download Hive from Apache servers
   ```
   $ wget https://dlcdn.apache.org/hive/hive-4.0.1/apache-hive-4.0.1-bin.tar.gz
   ```
2. Extract the downloaded archive
   ```
   $ sudo tar -xvf apache-hive-4.0.1-bin.tar.gz
   ```
3. Add a new environment variable and update PATH in `.bashrc`
   ```
   export HIVE_HOME=/opt/hive
   export PATH=$PATH:$HIVE_HOME/bin
   ```
4. Add HADOOP_HOME variable to `hive-env.sh`
   ```
   export HADOOP_HOME=/home/huser/hadoop-3.3.6
   ```
5. Edit Hive configuration file, i.e., `hive-site.xml`

   ```
   <xml version="1.0" encoding="UTF-8" standalone="no"?>
   <?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
   <configuration>
    <property>
     <name>javax.jdo.option.ConnectionURL</name>
     <value>jdbc:mysql://localhost/metastore_db?createDatabaseIfNotExist=true
           &allowPublicKeyRetrieval=true</value>
    </property>
    <property>
     <name>javax.jdo.option.ConnectionUserName</name>
     <value>hive</value>
    </property>
    <property>
     <name>javax.jdo.option.ConnectionPassword</name>
     <value>hive</value>
   ```

```
 </property>
 <property>
  <name>javax.jdo.option.ConnectionDriverName</name>
  <value>org.mariadb.jdbc.Driver</value>
 </property>
 <property>
  <name>hive.exec.local.scratchdir</name>
  <value>/tmp/hive</value>
 </property>
 <property>
  <name>hive.downloaded.resources.dir</name>
  <value>/tmp/hive_resources</value>
 </property>
 <property>
  <name>hive.server2.enable.doAs</name>
  <value>false</value>
 </property>
</configuration>
```

6. Create two directories in HDFS
   ```
   $ hadoop fs -mkdir -p /user/hive/warehouse
   $ hadoop fs -mkdir -p /tmp/hive
   ```
7. Change the permission of these directories
   ```
   $ hadoop fs -chmod 777 /user/hive/warehouse
   $ hadoop fs -chmod 777 /tmp/hive
   $ hadoop fs -chmod 777 /tmp
   ```
8. Install MySQL Java connector
   ```
   $ sudo apt install libmariadb-java
   ```
9. Copy the MySQL Java connector to Hive's lib folder
   ```
   $ cp /usr/share/java/mariadb-java-client.jar $HIVE_HOME/lib/mysql-connector-java.jar
   ```
10. Initialize the metastore schema
    ```
    $ schematool -initSchema -dbType mysql
    ```
11. Start Hive metastore and hiveserver2 services
    ```
    $ hive –service metastore &
    $ hive –service hiveserver2 &
    ```
12. Connect to Hive server on local machine via Beeline
    ```
    $ beeline -u jdbc:hive2://localhost:10000
    ```
13. Or Launch Hive CLI
    ```
    $ hive -u jdbc:hive2://localhost:10000
    ```

Note: It is highly recommended to write the code instead of copying and pasting.